

# Atoms and isotopes

Cover the answers. Answer each question out loud or on paper, then check. Tick the ones you know. Learn the 4 starred (★) questions first.

|         | QUESTION                                                               | ANSWER                                                                        | ✓                        |
|---------|------------------------------------------------------------------------|-------------------------------------------------------------------------------|--------------------------|
| 1       | State the approximate radius of an atom.                               | About 0.1 nm ( $1 \times 10^{-10}$ m)                                         | <input type="checkbox"/> |
| 2<br>★  | Describe the structure of an atom.                                     | A positive nucleus of protons and neutrons, surrounded by negative electrons. | <input type="checkbox"/> |
| 3<br>★  | State how the radius of a nucleus compares with the radius of an atom. | Less than 1/10 000 of the radius of the atom.                                 | <input type="checkbox"/> |
| 4       | State where most of the mass of an atom is concentrated.               | In the nucleus.                                                               | <input type="checkbox"/> |
| 5       | State the sign of the overall charge of the nucleus.                   | Positive                                                                      | <input type="checkbox"/> |
| 6       | State the relative charge of a proton.                                 | +1 (positive)                                                                 | <input type="checkbox"/> |
| 7       | State the relative charge of a neutron.                                | 0                                                                             | <input type="checkbox"/> |
| 8       | State the relative charge of an electron.                              | -1 (negative)                                                                 | <input type="checkbox"/> |
| 9       | State the relative mass of a proton.                                   | 1                                                                             | <input type="checkbox"/> |
| 10      | State the relative mass of a neutron.                                  | 1                                                                             | <input type="checkbox"/> |
| 11      | State the relative mass of an electron.                                | Very small (about 1/2000)                                                     | <input type="checkbox"/> |
| 12      | Describe the arrangement of electrons in an atom.                      | In energy levels at different distances from the nucleus.                     | <input type="checkbox"/> |
| 13<br>★ | State how an electron can move to a higher energy level.               | By absorbing electromagnetic radiation.                                       | <input type="checkbox"/> |
| 14<br>★ | State how an electron can move to a lower energy level.                | By emitting electromagnetic radiation.                                        | <input type="checkbox"/> |
| 15      | Name the two types of particle found in the nucleus of an atom.        | Protons and neutrons                                                          | <input type="checkbox"/> |